MadLoop with TIRust

One-loop reduction · registered campaign report

CutTools · Ninja · COLLIER · NinjaRS · TIRust

Starting MadLoop revision `9228fc58e8f8` · full helicity/color sums

Selected campaign: `invariant-symmetry-1`. Selected numerical identity: `f217762db032b8b7`.

Registered exporter anchor: `2c13246ffa63`. Observed formal paired-timing export revisions: not yet measured. Documentary revisions can differ while the registered exporter bytes remain identical.

Complete delivery qualification: incomplete. 11 of 11 physical cases have closed: 66 lane slots × 100 saved points. 11 complete cases; 11 pass 3e-6. Current numerical status: **passed**. The 1e-8 target is missed by 3 complete cases; all lane outcomes remain visible. The eleven-process numerical matrix, controls, existing-backend regressions, precision cohorts, implementation checks and unresolved stress cases retain separate statuses.

Precise performance acceptance: deferred to Hua-Sheng. Final timings will run on his local machine. Shared-host timings are provisional, without CPU reservation, and cannot establish the ≤1.05 upper confidence bound. This snapshot contains no new paired timing samples.

Current-source numerical checkpoint. This report selects `invariant-symmetry-1`. Earlier shared-host timings remain in the archived 19-page split-dp-terminal-1 report, bound to commit `584dae7875d2`. Those timings retain their historical source identity. Current performance gates remain deferred.

Runtime colour: green = faster, white = parity, red = slower. All displayed ratios are candidate / reference; smaller is faster.

Table 1: Main process matrix. Timings include the actual rescue work. Physics and runtime acceptance are separate.

Process · correction	Physics	Baseline µs/point	TIRust chain µs/point	Ratio · 95% interval	Existing-backend regressions
$u\bar{u} \to Z$ QCD · 2→1 · 100 points	pass	—	—	— deferred	deferred
$gg \to t\bar{t}$ QCD · 2→2 · 100 points	pass	—	—	— deferred	deferred
$u\bar{u} \to Zgg$ QCD · 2→3 · 100 points	pass	—	—	— deferred	deferred
$u\bar{u} \to ZZgg$ QCD · 2→4 · 100 points	pass	—	—	— deferred	deferred
$u\bar{d} \to W^+$ EW · 2→1 · 100 points	pass	—	—	— deferred	deferred
$e^+e^- \to \mu^+\mu^-$ EW · 2→2 · 100 points	pass	—	—	— deferred	deferred
$e^+e^- \to t\bar{t}\gamma$ EW · 2→3 · 100 points	pass	—	—	— deferred	deferred
$e^+e^- \to \mu^+\mu^-\gamma\gamma$ EW · 2→4 · 100 points	pass	—	—	— deferred	deferred

This study measures the new integration against the untouched starting branch, which already includes NinjaRS and earlier Fortran optimizations. It does not attribute those earlier improvements to TIRust.

1 Retained predecessor failures

The following qualified failures belong to earlier candidate implementations. They remain authoritative for those numerical sources and fixed policies. Results for the selected campaign appear in the main matrix; a new campaign does not replace or erase a previous candidate's failure.

Table 2: All retained qualified ancestor failures, including discovered and unselected attempts. Source identities and every read evidence file are bound in the report build manifest.

Process · comparison	Previous candidate source	Retained ratio · 95% interval	Retained failure evidence
$u\bar{u} \to Z$ `tirust-first`	`legacy` revision `c91cb85f3aa9` source SHA `74116e4bb2ba7abd`	1.12105 1.10349–1.13889 **fail**	`runs/qcd_n1_uux_z/tirust-first-attempt-3`

2 Five-backend numerical agreement

The same saved momenta, model parameters, coupling orders and scale are evaluated by all five tools. The regression criterion is `3e-6`, with `1e-8` as the ordinary-point target. Born, finite, simple-pole and double-pole coefficients are retained. Near-zero poles use the existing MadLoop comparison policy. Final accepted DP/QP outcomes and failed evaluations are retained; the histograms do not enumerate internal rescue attempts.

This table reparses the final complete steady warmup traversal: 100 saved points, baseline CutTools and five candidate reducers. It is untimed numerical evidence. It supplies neither paired runtime acceptance nor native-DP cohort timings. Internal retry counters and hardware FPE flags are **unobserved**. This warmup-only numerical traversal finishes before unit 6 is redirected, so its console diagnostics are retained. The numerical parser does not attribute or gate every console warning per call. Subsequent post-READY timed RUN batches suppress unit 6 output.

Process · saved points	CutTools	Ninja	COLLIER	NinjaRS	TIRust
$u\bar{u} \to Z$ main · 100 expected Baseline CT: **complete**	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 100 / QP 0 1e-8: pass
$gg \to t\bar{t}$ main · 100 expected Baseline CT: **complete**	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 100 / QP 0 1e-8: pass
$u\bar{u} \to Zgg$ main · 100 expected Baseline CT: **complete**	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 100 / QP 0 1e-8: miss	**pass** DP 100 / QP 0 1e-8: miss	**pass** DP 100 / QP 0 1e-8: miss	**pass** DP 87 / QP 13 1e-8: miss
$u\bar{u} \to ZZgg$ main · 100 expected Baseline CT: **complete**	**pass** DP 88 / QP 12 1e-8: pass	**pass** DP 97 / QP 3 1e-8: miss	**pass** DP 100 / QP 0 1e-8: miss	**pass** DP 96 / QP 4 1e-8: miss	**pass** DP 14 / QP 86 1e-8: miss
$u\bar{d} \to W^+$ main · 100 expected Baseline CT: **complete**	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 100 / QP 0 1e-8: pass
$e^+e^- \to \mu^+\mu^-$ main · 100 expected Baseline CT: **complete**	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 100 / QP 0 1e-8: pass
$e^+e^- \to t\bar{t}\gamma$ main · 100 expected Baseline CT: **complete**	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 100 / QP 0 1e-8: pass
$e^+e^- \to \mu^+\mu^-\gamma\gamma$ main · 100 expected Baseline CT: **complete**	**pass** DP 93 / QP 7 1e-8: pass	**pass** DP 100 / QP 0 1e-8: miss	**pass** DP 100 / QP 0 1e-8: miss	**pass** DP 100 / QP 0 1e-8: miss	**pass** DP 40 / QP 60 1e-8: miss
$u\bar{u} \to g\gamma$ control · 100 expected Baseline CT: **complete**	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 100 / QP 0 1e-8: pass
$gg \to H$ control · 100 expected Baseline CT: **complete**	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 100 / QP 0 1e-8: pass
$gg \to HH$ control · 100 expected Baseline CT: **complete**	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 100 / QP 0 1e-8: pass	**pass** DP 100 / QP 0 1e-8: pass

Every available comparison uses baseline CutTools rows on exactly matching point and split-order indices. A failed baseline leaves complete candidate lanes uncompared; their DP/QP counts remain observed. The separate 1e-8 target is not substituted for the 3e-6 regression criterion. Failed, incomplete and corrupt observations retain their diagnostics and never become successful rows.

Agreement is assessed using independent numerical backends and higher precision when a discrepancy occurs. Known difficult scalar-box regions are retained as stress cases. A backend's own success flag does not establish cross-backend agreement.

3 Completed physical pilot checks

These eight separately retained checks each evaluate 100 saved physical points with all five reducers. They use the common strict stability policy and pass the `3e-6` regression criterion. Their timings are ineligible for performance acceptance, and they do not fill the pending timing-campaign rows above. An independent audit reparsed 4,800 evaluations and checked the retained inputs, executable provenance and raw statuses.

> These historical pilots retain their original candidate sources in the pilot index. They do not establish numerical acceptance for the selected campaign `invariant-symmetry-1`.

Table 3: Historical physical pilots, independent of the runtime gate. Residuals exclude poles accepted by the documented small-pole exception. All four existing reducers finally accepted 100 DP results in each row.

Process · 100 points	Five-backend regression	Ordinary target	Largest symmetric residual	TIRust accepted DP / QP
$u\bar{u} \to Z$	**pass**	**pass**	2.69×10^{-13}	100 / 0
$gg \to t\bar{t}$	**pass**	**pass**	4.58×10^{-12}	100 / 0
$u\bar{u} \to Zgg$	**pass**	**miss**	2.81×10^{-6}	87 / 13
$u\bar{d} \to W^+$	**pass**	**pass**	1.95×10^{-12}	100 / 0
$e^+e^- \to \mu^+\mu^-$	**pass**	**pass**	2.65×10^{-12}	100 / 0
$u\bar{u} \to g\gamma$	**pass**	**pass**	2.61×10^{-11}	100 / 0
$gg \to H$	**pass**	**pass**	1.55×10^{-10}	100 / 0
$gg \to HH$	**pass**	**pass**	4.61×10^{-9}	100 / 0

Historical split-dp-terminal-1 QCD 2→3 adjudication. The original comparison against CutTools DP misses the `1e-8` target at 28 points. CutTools, Ninja and TIRust QP agree within `9.89e-11` on four diagnostic points. Across all 28 target misses, the saved TIRust and COLLIER results meet the target against CutTools QP; CutTools DP misses all 28, while Ninja and NinjaRS miss only point 39. Every comparison still passes the regression criterion. These fixed-QP diagnostics are explicitly unchecked and do not establish automatic rescue stability.

The index `experiments/numerical/completed-pilots-index.json` binds 491 retained files and preserves the original comparisons. The detailed QP adjudication is in `experiments/qcd-n3-adjudication/`. Final accepted precision does not enumerate every internal reduction attempt. The remaining three large-process pilots are outside this frozen eight-case index.

4 Absolute timing by reduction backend

Values below are warmed full-amplitude times in µs per point. Initialization, compilation and model generation are excluded and recorded separately. All tools use matching pole, helicity/color and stability settings. Cache behaviour and actual rescue work remain part of the measurement. Recorded backend/precision histograms identify the final accepted reducer for every timed amplitude; they do not count every internal rescue attempt. Control-process speeds are descriptive; the ≤1.05 runtime gate applies to the eight main processes.

Table 4: Per-backend timing observations. Missing measurements are not represented as zero.

Process	CutTools	Ninja	COLLIER	NinjaRS	TIRust
$u\bar{u} \to Z$	—	—	—	—	—
$gg \to t\bar{t}$	—	—	—	—	—
$u\bar{u} \to Zgg$	—	—	—	—	—
$u\bar{u} \to ZZgg$	—	—	—	—	—
$u\bar{d} \to W^+$	—	—	—	—	—
$e^+e^- \to \mu^+\mu^-$	—	—	—	—	—
$e^+e^- \to t\bar{t}\gamma$	—	—	—	—	—
$e^+e^- \to \mu^+\mu^-\gamma\gamma$	—	—	—	—	—
$u\bar{u} \to g\gamma$	—	—	—	—	—
$gg \to H$	—	—	—	—	—
$gg \to HH$	—	—	—	—	—

5 Initialization and accepted precision

Each cell gives the first-amplitude initialization plus complete-stream warmup time in seconds, followed by the fraction of timed amplitudes finally accepted in QP. Process startup and parameter-card loading are outside the cold interval. A DP-only label refers to accepted amplitudes; it does not assert that no unsuccessful internal attempt occurred. The warmed timings on the previous page include all rescue work.

Table 5: Cold intervals and final accepted precision. Exact backend/precision counts and point-level results accompany the raw observations.

Process	CutTools	Ninja	COLLIER	NinjaRS	TIRust
$u\bar{u} \to Z$	—	—	—	—	—
$gg \to t\bar{t}$	—	—	—	—	—
$u\bar{u} \to Zgg$	—	—	—	—	—
$u\bar{u} \to ZZgg$	—	—	—	—	—
$u\bar{d} \to W^+$	—	—	—	—	—
$e^+e^- \to \mu^+\mu^-$	—	—	—	—	—
$e^+e^- \to t\bar{t}\gamma$	—	—	—	—	—
$e^+e^- \to \mu^+\mu^-\gamma\gamma$	—	—	—	—	—
$u\bar{u} \to g\gamma$	—	—	—	—	—
$gg \to H$	—	—	—	—	—
$gg \to HH$	—	—	—	—	—

6 Full versus poles-only tensors

10 of 12 predeclared settings completed seven alternating full/poles-only timing pairs; two DP settings were refused in both modes. All returned requested pole coefficients agree. These warmed tensor diagnostics use the historical release, with numerical caches disabled. They do not qualify full-amplitude performance.

Table 6: Median wall µs per call and geometric mean paired ratios. Green favors poles-only; red favors full evaluation. Intervals use seven paired log ratios and Student-t with six degrees of freedom.

Tensor · N / rank	Route	Prec.	Full µs	Poles µs	Poles / full · 95% interval
Massive tadpole N=1 · rank 6	Public Fortran	DP	9.01	9.42	**×1.041** 1.032–1.049
Massive tadpole N=1 · rank 6	Public Fortran	QP	1828	1838	**×1.005** 0.999–1.011
Massive bubble N=2 · rank 4	Generated adapter	DP	11.8	11.3	**×0.951** 0.939–0.964
Massive bubble N=2 · rank 4	Generated adapter	QP	1738	1121	**×0.646** 0.644–0.649
Infrared triangle N=3 · rank 3	Generated adapter	DP	8.95	8.78	**×0.977** 0.954–1.000
Infrared triangle N=3 · rank 3	Generated adapter	QP	1038	858	**×0.823** 0.815–0.832
Massive box N=4 · rank 4	Generated adapter	DP	32.2	18.0	**×0.558** 0.551–0.564
Massive box N=4 · rank 4	Generated adapter	QP	15384	2220	**×0.144** 0.141–0.146
Massive pentagon N=5 · rank 2	Generated adapter	DP	—	—	Both modes refused
Massive pentagon N=5 · rank 2	Generated adapter	QP	39078	2177	**×0.056** 0.055–0.057
Massive hexagon N=6 · rank 2	Generated adapter	DP	—	—	Both modes refused
Massive hexagon N=6 · rank 2	Generated adapter	QP	84852	5144	**×0.061** 0.060–0.061

Scope and observed limits. The rank-six tadpole calls the public Fortran bridge; the other tensors use the existing generated QCD 2⟶4 adapter, whose rank capacity is five. No generated rank limit was changed. The DP tadpole is about 4% slower in poles-only mode; QP is near parity. The massive N=5 and N=6 fixtures have zero poles, so their large QP gains do not describe infrared-singular pentagons or hexagons. DP refusals carry no timing claim.

Numerical caches are off; structural plans and scratch remain warm. Each measured batch lasted 0.825–1.925 s on pinned CPU 9. Driver tracing and coefficient output are outside the timer; input restoration, adapter mapping and bridge serialization remain inside. Shared-host compilation was active; host-load snapshots and CPU times are retained. These timings are explicitly ineligible for the full-amplitude acceptance gate.

Requested poles match within the declared DP/QP bounds; the largest scaled residual is `3.47e-18`. Omitted finite slots are NaN in the generated adapter and zero placeholders in the public bridge. The historical DSO is pinned by SHA256 `49af12f0b882…cd71ca12` and subsequent builds are outside this measurement. Raw batches, exact identities, and the independently audited 79-file inventory are retained under `experiments/poles-timing/`.

7 Methods and qualification

7.1 Physical samples and timing protocol

The main matrix contains QCD and electroweak corrections at every multiplicity from 2⟶1 through 2⟶4, using `loop_sm` and `loop_qcd_qed_sm_Gmu`. Each main case declares 100 deterministic points. On-shell 2⟶1 points vary their boost at the physical invariant; higher multiplicities use 1 TeV. Mixed QCD/QED and loop-induced cases supplement the main matrix. Threshold, soft/collinear, small-Gram and complex-mass cases are separate stress controls.

Performance acceptance uses at least seven alternating paired repetitions. The minimum batch target is one second; calibration aims for two seconds to accommodate shared-host speed variation. Actual durations are retained, and every batch must reach 95% of the minimum. The starting revision and candidate share their compiler, library installations, model/cards, point order and physical settings. Timing begins after the campaign's builds finish and uses one allowed CPU. The upper 95% paired ratio bound must be ≤1.05 separately for every main process. Averages cannot hide a failing row. A FULL–SKIP quantity, when reported in supporting data, is an **estimated reduction stage** rather than a pure tensor-kernel stopwatch.

All timing attempts are retained. Only an objectively short series with passing numerical checks can be retried within the same candidate campaign. A qualified runtime failure remains authoritative; a later favorable repetition cannot erase it. Source optimizations require a new candidate identity and campaign.

The loop-induced controls retain their generated default chain `7|6|1` their prescribed TIRust-first chain is `9|7|6|1`. These are configured chains, not measurements of which reducer ultimately supplied a result.

The recorded cold interval includes first-amplitude initialization and warming the complete saved point stream; it excludes process startup and parameter-card loading. Numerical objects and linked archives are bound to the retained compiler recipes and hashes. The benchmark runs on a shared host; host-load observations accompany timing pairs, and this study does not claim exclusive use of the machine.

The common benchmark card sets `MLStabThres=1e-6` for both revisions and all reducers, with independent stability checks enabled. This policy was frozen before acceptance measurements after a native stress audit exposed correlated errors below the original threshold. Production retains its `1e-3` default. The tighter estimator does not replace the independent numerical truth gate or guarantee detection of every error.

7.2 Interface, poles and caches

MadLoop selects TIRust explicitly as `MLReductionLib=9`. A Fortran adapter transfers supplied invariants, complex squared masses and the squared renormalization scale to a separate Rust bridge. Tensor moments are permuted into MadLoop order and contracted in Fortran. The quadruple-precision path preserves binary128 inputs and uses MPFR working precision; double-double arithmetic is not labelled quadruple precision.

`LoopPoleMode` selects bare-loop UV/IR residues while preserving the ordinary finite coefficient. The distinct poles-only diagnostic path omits finite scalar roots and finite tensor arithmetic; it preserves double-pole contributions to simple poles and scaleless UV/IR separation. Partial results cannot be used as complete physical amplitudes. Cache and precision stores are separated by evaluation mode, and stability rechecks use independent values.

> Detailed interface, cache, QP, poles-only, and relocation qualification records have not yet been incorporated in this data release. Their absence is not a passing result.

7.3 Earlier measured Fortran change

The selected optimization computes ordered momentum prefixes once, retaining the original zero seed and addition order. For six propagators it reduces prefix additions from 60 to 20. At each of `-O0` and `-O3`, 481,536 comparisons passed bitwise output and IEEE-exception checks across DP, QP and four rounding modes. Isolated six-propagator kernel median paired ratios were 0.443/0.347 (DP/QP, `-O0`) and 0.703/0.356 (`-O3`). These are shared-host microbenchmarks, not physical-amplitude speedups. Full-wrapper measurements were smaller and less consistent. Raw observations and source are retained in `experiments/fortran-prefix/`.

7.4 Retained numerical limitations

The bridge's 66 DP/QP stress evaluations include 14 explicit DP refusals and one served DP tensor coefficient above the diagnostic `3e-6` criterion. All 33 QP evaluations pass. For the failing massive rank-five box, native MadLoop reversal checks at the default `MLStabThres=1e-3` can accept correlated DP errors. At `1e-6`, the actual production rescue block with real tensor callbacks switches to QP and resolves the discrepancy below `1.2e-15`. The fixture, failing components,

native check values and QP adjudication remain under `qualification/tirust-bridge/stability-audit/`. This isolated control-flow test omits physical helicity/filter initialization and does not replace matrix-element comparisons.

Pole-only tests prove that finite evaluators and finite tensor arithmetic are skipped. This does not imply every scalar topology is faster: illustrative warm scalar measurements include slower simple massless triangles and boxes, alongside faster massive and soft examples. A completely collapsed massless scalar box returns an explicit error. No unsupported request is counted as accelerated execution.

7.5 Reproduction and provenance

The accompanying `campaigns/invariant-symmetry-1/cases.json` fixes the matrix and thresholds; `campaigns/invariant-symmetry-1/summary.json` is derived from retained observations. The selected manifest is `campaigns/invariant-symmetry-1/manifest.json`. Its ancestor summary snapshots and qualified failures retain the previous candidates' identities separately. The source plan, baseline manifest, dependency revisions, build environment, and raw measurement paths accompany this report. Rebuilding the PDF does not rerun numerical measurements.

Component	Recorded environment
Baseline	`9228fc58e8f8e48dadc70a6fcfb6617845aaeb1f`
Rust	rustc 1.97.1 (8bab26f4f 2026-07-14) (built from a source tarball)
Fortran	GNU Fortran (GCC) 15.2.0
Typst	typst 0.15.1 (9dfd3a08)
TIRust	`854b79dcb7fffcb35e92b4a8a3dd443369b798c4`
oneloopmaster	`bd703d09a78f7849a6b86a07096e6f222fc01193`

Presentation follows the pyAmpliCol report at revision 7266c1600f6e: its page geometry, fonts, navy headings, compact tables and logarithmic timing palette. All numerical measurements and conclusions in this report belong to the present MadLoop integration.

8 Timing eligibility and control-process scope

TODO — Hua-Sheng, local machine. Rebuild the delivered candidate and baseline with identical pinned inputs; replay all 100 saved points; then run at least seven alternating single-core pairs with approximately one-second batches. Report initialization and actual backend/precision use separately. The upper 95% ratio bound ≤1.05 must be checked for every main chain and existing-backend comparison. Shared-host work is best effort without reserving CPUs; it cannot pass this final performance requirement.

User amendment: `PLAN-AMENDMENT-2026-11-10.md`. Runnable instructions: Hua-Sheng's local precise timing handoff. It covers fresh installation, local registration, full numerical replay, paired timing and a separate local PDF; the handoff is not a measurement.

Additional local accepted-precision diagnostics: full-stream precision TODO. This separate diagnostic cannot satisfy the formal 1.05 runtime gate.

Formal main-process timing uses a 2 s calibration target and the frozen 0.95 s minimum observed batch duration. Each batch traverses all 100 saved points. All valid attempts remain visible; a qualified poor result cannot be replaced by a favorable retry. These limits are distinct from the separate traced-cohort diagnostic duration rule.

Main process	Min. baseline s	Min. candidate s	Retained paired attempts
$u\bar{u} \to Z$	—	—	0 · 0 short
$gg \to t\bar{t}$	—	—	0 · 0 short
$u\bar{u} \to Zgg$	—	—	0 · 0 short
$u\bar{u} \to ZZgg$	—	—	0 · 0 short
$u\bar{d} \to W^+$	—	—	0 · 0 short
$e^+e^- \to \mu^+\mu^-$	—	—	0 · 0 short
$e^+e^- \to t\bar{t}\gamma$	—	—	0 · 0 short
$e^+e^- \to \mu^+\mu^-\gamma\gamma$	—	—	0 · 0 short

Minima include all successfully reparsed retained timing attempts and comparison labels, including ineligible short attempts. Missing or invalid observations do not supply a duration. Selected runs and each attempt's exact minima, labels and eligibility remain in the dossier.

8.1 Controls in the formal campaign

Mixed QCD/QED and loop-induced controls require numerical qualification. Their speed measurements are descriptive and do not carry the main-process ≤1.05 runtime gate. Generic raw-summary failure labels can include slow controls; the role-specific interpretation below preserves those labels without turning control speed into a main acceptance failure.

Control	Formal paired numerical evidence	Raw summary status	Runtime gate
$u\bar{u} \to g\gamma$	**pending**	pending	not applicable
$gg \to H$	**pending**	pending	not applicable
$gg \to HH$	**pending**	pending	not applicable

The untimed numerical and provisional shared-host tables are separate datasets. Their completed observations do not populate these formal-campaign fields.

9 Native-DP and rescued cohort diagnostics

Traced full-amplitude subset diagnostics. DP includes internal method selection; compensated DP and naturally rescued QP are separate. Per-call tracing adds overhead. Different frozen subsets cannot be used as backend speed ratios. The diagnostic 90% duration rule is distinct from the main paired 95% rule.

Process	CutTools	Ninja	COLLIER	NinjaRS	TIRust
$u\bar{u} \to Z$	pending	pending	pending	pending	pending
$gg \to t\bar{t}$	pending	pending	pending	pending	pending
$u\bar{u} \to Zgg$	pending	pending	pending	pending	pending
$u\bar{u} \to ZZgg$	pending	pending	pending	pending	pending
$u\bar{d} \to W^+$	pending	pending	pending	pending	pending
$e^+e^- \to \mu^+\mu^-$	pending	pending	pending	pending	pending
$e^+e^- \to t\bar{t}\gamma$	pending	pending	pending	pending	pending
$e^+e^- \to \mu^+\mu^-\gamma\gamma$	pending	pending	pending	pending	pending
$u\bar{u} \to g\gamma$	pending	pending	pending	pending	pending
$gg \to H$	pending	pending	pending	pending	pending
$gg \to HH$	pending	pending	pending	pending	pending

Cells give frozen point counts and diagnostic median µs per call. Empty cohorts have no timing. Short batches, migrations or failure flags cannot yield a valid median. The complete original stream is warmed before membership is frozen; QP timings include failed DP work and natural rescue.

10 Registered implementation and delivery status

Required criterion	Observed status
all eleven five backend numerical	**passed**
eight main runtime gates	**deferred to Hua-Sheng**
existing backend regressions	**deferred to Hua-Sheng**
Installed ABI / Fortran: recorded test groups	**passed**
installer metadata hardening	**passed**
native dp and rescued cohorts	**pending**
paired campaign overall	**deferred to Hua-Sheng**
Current statistics round trip	**passed**
stress accuracy	**open**
Subprocess/event cache lifecycle	**passed**

Installed package. DSO `eb3d67fdc99b742b` Fortran archive `0c86349a4fd0dadd` actual installer source `059bb9c97919`. This is distinct from the registered exporter anchor and each observed export HEAD.

The public Fortran parity replay contains 65 calls per installation. The raw DP replay contains 90 calls per library, with 80 successes and 10 expected failures. Cache transport rejection is checked against the retained old-library negative control. These checks qualify the recorded interface and tested transport/cache rejection behavior; their elapsed times are not physical-amplitude benchmark samples.

Integration work remains open. Selected standalone runtime, control and diagnostic proofs are recorded below. Adaptive-bridge production promotion remains pending a separately bound receipt. The installer metadata correction has a bound production test receipt. These scoped proofs do not close the full campaign, stress or performance gates.

Subprocess/event cache lifecycle: passed. Four fresh workers, 76 checked native-DP physical calls and 96 separate DP/QP tensor-key requests; exact cache/observer parity. Physical-QP rescues, rank-six, EW, poles-only and runtime remain outside this lifecycle scope.

`experiments/subprocess-cache-lifecycle-current-qualification-2/qualification-summary.json`

Current statistics: passed. One retained invocation passed 10 tests and four compiled synthetic Fortran tally/writer subtests, using current production Python readers. Legacy aliases, TIRust/NinjaRS precision counters and aggregation are covered. This qualifies the identified statistics sources; it neither recovers the historical unbound 92-pass invocation nor establishes amplitude, native-reducer or performance results.

`experiments/statistics-roundtrip-current-1/manifest.json`

Higher-rank terminal Fortran qualification is bound to the completed source, IEEE/trap and fixed-input coefficient checks in `experiments/dp-higher-rank-production/validation-summary.json`. Its compiled checks and CutTools parity do not substitute for the registered five-backend campaign or establish runtime admission frequency.

11 Retained accuracy limitations and diagnostics

The report binds the completed portable inventories and their recorded scope; it does not rerun scientific adjudication. Historical library results and private prototypes do not qualify the current installed library. Ordinary numerical and paired runtime gates remain separate. The historical split-dp-terminal-1 QCD n3 optional target and higher-precision workstream are diagnostic disclosures; neither adds a stress acceptance requirement.

Case	Status	Qualification limit
EW threshold mass transport	**open**	Historical diagnostic retained by split-dp-terminal-1: Historical 49af, private mass-transport prototype. QP squaring of existing MP masses reduces the recorded group-5 Gram discrepancy from 1.018% to 1.60e-20, while both arms still refuse 37 threshold tensor parents. Status 120 and accuracy −1 are unchecked; the TIRust sum is incomplete. Matching successful scalar/rank-one subsets does not qualify the failed tensors, full amplitude or performance. `experiments/ew-qp-mass-prototype/observer-native-design/attempt2/execution-1/geometry-audit/README.md`
QCD soft/collinear prototype	**open**	Historical diagnostic retained by split-dp-terminal-1: Historical 49af, six private replays. Topology correction plus Born evaluation at repaired coordinates gives NinjaRS status 322 and a 1.33e-15 analytic double-pole error; topology-only status 322 still misses 3e-6. Finite/single-pole truth remains unavailable. TIRust retains 51 nonsingular conditioning refusals and failed status 420; its incomplete sums are not amplitudes accepted for physics. `experiments/qcd-topology-rescue-native/RESULTS.md`
Default stability estimator	**open**	Historical split-dp-terminal-1 disclosure, same installed eb3d bridge: its completed replay confirms four inaccurate DP acceptances at the production 1e-3 default (maximum finite symmetric error 7.03e-4) and four QP rescues at 1e-6 (maximum symmetric error 5.97e-16). This synthetic host-helper fixture does not qualify physical amplitudes; the registered policy remains 1e-6. `experiments/default-stability-current-1/README.md`
QCD n3 ordinary target	**open**	Historical diagnostic retained by split-dp-terminal-1: Current eb3d campaign: all 100 points pass 3e-6, while the same 28 finite-term 1e-8 misses against CutTools remain. Retained independent QP comparisons support a CutTools DP reference bias and a smaller Ninja/NinjaRS discrepancy at point 39; a reducer cluster alone is not truth. Those historical fixed-QP results are unchecked status 120, not an automatic-rescue certificate. Original comparisons and acceptance gates are unchanged. `campaigns/split-dp-terminal-1/experiments/qcd-n3-target-triage-1/README.md`
Complex-mass extremes	**open**	Historical diagnostic retained by split-dp-terminal-1: Nineteen moderate points agree; original extreme automatic-policy points remain unqualified. Private corrections are unchecked diagnostics. `experiments/complex-mass/README.md`
QCD higher precision	**diagnostic**	Historical split-dp-terminal-1 disclosure: a separate private all-512-bit replay completed 232 tensor calls with actual runtime mapping and exact ordinary-point parity. Forty-six contracted tensor slots retain discrepancies despite close final projections. The changed precision and admission gate do not qualify the adaptive 256-to-512 library, general physical accuracy or runtime acceptance. `experiments/qcd-topology-rescue-higher-precision-native2/README.md`

The historical c91 qualified runtime failure remains an ancestor result with its original source and interval. New source candidates do not erase it. Ordinary-point agreement and a passing eight-process runtime gate alone cannot declare the full integration goal complete.

12 Additional implementation proofs

Each updated claim below is derived from an explicitly selected package whose index and retained files are hash-bound to this dossier. These separate experiments do not replace the source identity, precision usage or numerical observations of the ordinary campaign.

The experimental bridge is an isolated development build, whose source and evidence are recorded in this public feature branch. It has not replaced the installed TIRust bridge used for the ordinary campaign. The branch is `madloop_tirust` the intended future PR target is `madloop_ninjars`.

12.1 Historical split-dp-terminal-1 QCD n4 target misses

The archived split-dp-terminal-1 report retains 45 distinct target-miss points and its old DP/QP counts. Those observations belong to that historical source. Current 1e-8 target outcomes are derived only from this report's closed current checkpoint.

`history/report-before-invariant-symmetry-qualification/archive-manifest.json`

12.2 Experimental adaptive-precision bridge

Separately observed Fortran runtimes now cover debug: 13 genuine binary128 components, 10 adaptive and 1 legacy tensor calls; release: 13 genuine binary128 components, 10 adaptive and 1 legacy tensor calls. Actual 256/512-bit dispatch, cache and check telemetry are retained, including the deliberate error call. These are standalone tensor/transport proofs. Accepted amplitude precision remains unobserved; full-amplitude accuracy, Born behavior and performance are not qualified. Production promotion remains pending a separate receipt.

`experiments/adaptive-qp-fortran-qualification/runtime1/evidence-index.json; experiments/adaptive-qp-release-fortran-qualification-v2/runtime1/evidence-index.json`

12.3 Born/effective-point consistency

Actual backends 8 and 9 complete 8 ordinary and 16 fixed-QP real-Z controls. The 24 retained logs contain 0 W03 warnings; 48 fixed-QP observations carry unchecked status 120. The requested Born/coordinate views and recorded coefficient comparisons pass within this control scope. Unchecked calls do not establish independent stability. CMS/stress coverage, a full campaign and performance remain unqualified.

`experiments/born-effective-point-native89-qualification-1/evidence-index.json`

12.4 Symmetric invariants: component proof

The v2 extracted components pass 5376 DP/QP pairs and 48 trap pairs; 2 deliberate mutations are detected. Caller words, IEEE state, halting and rounding safeguards are retained. This component proof alone does not qualify a physical amplitude.

`experiments/ew-invariant-symmetry-prototype/v2/fortran-fixtures/qualification-1/index.json`

12.5 Symmetric invariants: separate EW stream

The v2 candidate accepts 100 DP / 0 QP points on one frozen 100-point EW stream, versus 82 DP / 18 QP for its old-source comparator. All 6 retained references pass 3e-6 and 1e-8; The retained console has zero W03 lines, and this instrumented diagnostic records zero IEEE state differences. Muted ordinary final-traversal diagnostics are not inferred from those console counts. These observations belong to the separate v2 candidate. They do not replace any of the older campaign's 66 numerical lanes or its provisional timing identity, and supply no runtime acceptance.

`experiments/ew-invariant-symmetry-prototype/v2/runtime-1/qualification-1/index.json`

12.6 Positive-width CMS diagnostic remains open

The 15 new positive-width rows retain 6 DP refusals, 6 returned QP tensors and 3 returned COLLIER tensors. All 15 rows retain IEEE flags; the combined analysis preserves 191 open items, including ancestral controls. Protocol completion is not CMS acceptance.

`experiments/generated-rank6-poles-proposal/attempt1/cms-width1/runtime1/index.json`

12.7 Common-input tensor diagnostic remains open

All 30 synthetic rows have exact effective inputs. The retained comparison still reports 3 comparison discrepancies, 6 refusals and 18 host IEEE observations. These are retained comparator observations, not an attribution of fault to a reducer. No physical or performance qualification follows.

`experiments/generated-rank6-poles-proposal/attempt1/common-input-proposal/runtime1/index.json`

12.8 Fixed-QP mass transport remains open

Nonmass input comparisons pass in the retained transport diagnostic. TIRust refusal counts by its two saved points are [0, 37] for the original arm and [0, 37] for the candidate arm. QP exception observations remain retained. Threshold aggregates are incomplete; checked-QP, physical accuracy and performance remain unqualified.

`experiments/ew-qp-mass-prototype/observer-native-design/attempt2/execution-1/evidence-index.json`

12.9 Installer metadata hardening

The strict JSON version-type correction is applied to production installer source 4397d9f60e146354. The bound receipt records 27 passing tests and 37 passing subtests, with the existing qualified prefix still available and QP-capable. Receipt provenance: Transcribed from actual completed tools session52323; no raw pytest log captured. This closes the selected metadata-hardening item; it does not supply new native or numerical qualification.

`experiments/installer-metadata-promotion-1/index.json`

12.10 Symmetric invariants: Fortran template promoted

The applied Fortran template is 4ccd4456d0b28f39. Its bound source-comparison receipt records 316 matching canonical statements in the qualified EW adapter context. All 4 tests pass: 3 compiled fixture methods and 1 Python permutation method. No real reducer or new physical replay is invoked by these tests. New exports can use this template; the older campaign's 66 lanes and provisional timings remain tied to their original source. Adaptive-bridge production promotion and final performance acceptance remain separate.

`experiments/ew-invariant-symmetry-prototype/v2/promotion-qualification-1/index.json`

12.11 Higher-rank finite residual clarification

The additive audit separates the worst finite symmetric residual, approximately 0.001328503784, from the aggregate maximum 1 in the simple-pole slot. The aggregate value includes near-zero UV comparison noise; it is not the finite residual. The original diagnostic remains preserved. Finite discrepancies and the comparator's requested-accuracy observations remain open; this offline audit establishes neither a reducer fault nor physical accuracy or performance.

`experiments/generated-rank6-poles-proposal/attempt1/common-input-proposal/finite-audit-1/index.json`

12.12 Adaptive fixed-QP private replay

The completed private replay serves 400 calls, recovers 37 former refusals, preserves 363 formerly served outputs bitwise, and changes W03 lines from 74 to zero. Both public statuses remain unchecked fixed-mode 120. The additive offline classification correction retains the original controller's complete=false failure and makes no native retry. Successful service establishes neither independent truth nor checked stability or per-call 512-bit execution. This is retained synthetic/private diagnostic evidence, separate from the current campaign's installed bridge. Physical amplitudes, general stress, production promotion and performance remain unqualified; recorded wall durations are administrative.

`experiments/adaptive-qp-physical-replay-qualification-1/evidence-index.json`

12.13 Adaptive replay: independent backend8 audit

Backend8 final sums agree with the private adaptive replay at maximum symmetric residuals at most 2.685e-16 (ordinary) and 2.029e-16 (threshold). Both serve 400 calls with the same physical/Born inputs. All 200 threshold native scales differ: backend8 squares in DP before promotion, backend9 after promotion. This is not identical native input data or checked truth; fixed-mode status120 remains unchecked. This is retained synthetic/private diagnostic evidence, separate from the current campaign's installed bridge. Physical amplitudes, general stress, production promotion and performance remain unqualified; recorded wall durations are administrative.

`experiments/adaptive-qp-physical-replay-independent-8-audit-1/evidence-index.json`

12.14 Adaptive masked checks and retained trapped failure

Masked adaptive threshold output passes independent automatic checks with status323: 600 QP calls served, 489 former served outputs preserved bitwise and 111 former refusals recovered across observed modes [4,5,4]. Ordinary status219 is accepted in DP and adds no QP coverage. Both original and adaptive trapped workers SIGFPE during threshold DP warmup2; the combined trap gate remains false. Offline evidence localizes the shared DP Cayley DIVSD instruction; the precise hardware exception remains inferred. Explicit rotation identity and per-call precision remain unavailable. This is retained synthetic/private diagnostic evidence, separate from the current campaign's installed bridge. Physical amplitudes, general stress, production promotion and performance remain unqualified; recorded wall durations are administrative.

`experiments/adaptive-qp-checked-trap-proposal-1/qualification-1/evidence-index.json`

12.15 Rank-six Feynman parameters: all-component fixture cross-check

The later independent integration corroborates all 210 finite components (420 parts) of the same synthetic fixture: 420 point-estimate passes at 3e-6 and 1e-8, maximum TIRust residual at most 9.625e-20. COLLIER passes 298/420 at 3e-6 and 92/420 at 1e-8; original answers and flag -1 remain retained. Analytic error alone preserves 420 TIRust passes at 3e-6; at 1e-8 it gives 357 agreements and 63 inconclusive parts. The vector arithmetic has a 64-bit significand. Analytic bounds exclude finite node/weight/arithmetic error; convergence supports a practical cross-check, without a whole-program certificate or cache-bug attribution. This is retained synthetic/private diagnostic evidence, separate from the current campaign's installed bridge. Physical amplitudes, general stress, production promotion and performance remain unqualified; recorded wall durations are administrative.

`experiments/rank6-feynman-parameter-all-components-qualification-1/index.json`

12.16 Rank-six Feynman parameters: seven-component cross-check

Independent direct integration corroborates 7 sampled finite components (14 real/imaginary parts); maximum observed TIRust residual at most 4.762e-17. Five of six disputed COLLIER parts miss 3e-6, consistent with retained flag -1. The analytic-error-only maximum upper residual is at most 3.558e-7. Seven samples do not qualify all 210 components. Finite node/weight/arithmetic error is checked empirically; no whole-program certificate or COLLIER cache-bug attribution is supplied. This is retained synthetic/private diagnostic evidence, separate from the current campaign's installed bridge. Physical amplitudes, general stress, production promotion and performance remain unqualified; recorded wall durations are administrative.

`experiments/rank6-feynman-parameter-qualification-1/index.json`

12.17 Guarded adaptive bridge: closed private compilation

The guarded private release build completes with four newly compiled crates and 140 reused dependency artifacts. All 1,193 protected bindings remain unchanged; 499 source identities and 133 retained code texts bind the sole reviewed signed-zero denominator guard. The new library 69722452c234f494 is distinct from the retained old private c5de24e6c41a90ca library and the campaign's eb3d67fdc99b742b bridge. Both prior failed builds remain retained: LLVM allocation failure under a 4 GiB address-space limit, then a separate 600-second deadline. The Cargo-banner verifier correction is retained without another build or DSO read. This package qualifies compilation and source provenance only. The new library was not loaded, installed or numerically exercised in this build package; physical accuracy, precision, stability, trap behavior, production promotion and runtime performance remain unqualified here. Recorded build durations are administrative and supply no performance claim.

`experiments/cayley-guarded-adaptive-build-qualification-1/evidence-index.json`

12.18 Guarded adaptive physical replay: masked equality, new trapped failures

The guarded private bridge preserves both closed adaptive masked references bitwise. Checked mode accepts the ordinary point in DP (status219) and the threshold point in QP (status323), with 600 QP calls in modes [4,5,4]. Forced masked QP serves 400 calls; status120 and accuracy -1 remain unchecked. The original-backend comparison remains a nonreference diagnostic: its permissive regression passes despite maximum relative residual 0.9241097961, nonidentical outputs and the retained threshold status420 with 111 refusals. Both guarded trapped workers SIGFPE before observer journals; their FileNotFoundError assessments remain retained. All joint numerical, host-stability, QP-coverage and trap gates remain false. Offline localization identifies other DIVSD sites in DP gram_indicator and multiprecision dilogarithm series; the actual operands and exact exception subtype remain unknown. The 69722452 guarded library, old c5de24e6

private library and campaign eb3d67fd bridge remain distinct. Two optional verifier-test policy failures are retained separately from the passing portable replay and physical outcomes. This two-point private diagnostic establishes neither independent physical truth, explicit rotation identity, per-call working precision, production promotion nor runtime performance. Recorded execution duration is administrative; precise timing remains a local TODO.

`experiments/cayley-guarded-adaptive-physical-qualification-1/evidence-index.json`

12.19 Current CMS coverage

incomplete: 6 expected automatic requests. This selected closed capsule records 6 amplitude streams started; 6 completed. Each CMS final stream requires 21 points and 42 order outcomes; warmup, bootstrap and final phases remain separate. Integrity/preparation/preflight do not establish checked full-stream agreement. Failed, unexecuted and unchecked outcomes remain retained. No repaired-geometry EW oracle, broad stress/CMS acceptance or performance qualification follows. Native orchestration completed 6 requests, while 0 complete streams were host checked and 252 final order outcomes remain unexecuted. Individual recorded warmup acceptance does not qualify a request or imply READY/RUN/final traversal. Worker launches, observed amplitude entry, bootstrap rows and repeated warmup observations do not determine the hidden amplitude evaluation count. Preparation/preflight are inherited from explicitly indexed prior setup; this capsule does not claim fresh CLI exports or builds. Explicit failed ancestor capsules and all their request rows remain retained separately; successor observations do not erase them.

`experiments/current-cms-replay-retention-2/evidence-index.json`

12.20 Current physical stress coverage

failed: 24 expected automatic requests. This selected closed capsule records 4 amplitude streams started; 4 completed. Each stress stream has two saved points. Integrity/preparation/preflight do not establish checked full-stream agreement. Failed, unexecuted and unchecked outcomes remain retained. No repaired-geometry EW oracle, broad stress/CMS acceptance or performance qualification follows. 4 host-checked two-point streams supply 8 qualified final point evaluations and 12 observed warmup calls; these observations do not count hidden evaluations. 5 of 6 checked-subset comparisons fail the 3e-6 criterion. Complete six-lane coverage and agreement remain separate gates. The retained comparator's ordinary_target_passed field applies the 1e-8 target to the whole two-point stream. Deferred and not-started requests remain explicit; the stage outcome is not replaced by a passing subset. The stage ended with a lock timeout before backend 8 launched. Four threshold streams were host checked; five of their six pairwise comparisons fail the regression criterion. One stream was deferred and 19 were not started. Preparation/preflight are inherited from explicitly indexed prior setup; this capsule does not claim fresh CLI exports or builds.

`experiments/physical-stress-current-portable-4-terminal-2/evidence-index.json`

12.21 Closed current QCD n3 target investigation

The bounded offline investigation retains the same 1,200 rows and 4,800 coefficient values as the prior closed result, including the same 28 finite-only 1e-8 target-miss points. All Born and pole comparisons pass 1e-8. The largest CutTools/TIRust symmetric finite residual is approximately 2.806420647e-6 at point 93; the largest non-CutTools pair residual is approximately 8.595156396e-8 at point 39. Cross-reducer consistency supplies no independent truth claim. This seven-input investigation runs no new native or QP evaluation and cannot replace the main full11 checkpoint or change any qualification or performance gate.

`experiments/invariant-symmetry-qcd-n3-target-analysis/portable-analysis2/index.json`

Actual Fortran tensor transport, unchecked fixed-QP controls and a separate physical stream each have a defined scope. They do not establish adaptive full-amplitude accuracy, general stress qualification or final runtime acceptance. The older campaign remains unchanged.